

RMIT UNIVERSITY

Assessment 2 — Technical Report



COSC2671

Social Media and Network Analysis

Tamil Nadu Election 2026 Public Opinion and Network Analysis

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1. Introduction

Social media platforms have become central spaces for political communication, public interaction, and opinion formation during modern elections. Political parties increasingly rely on digital media to distribute campaign narratives, influence voter perception, and maintain continuous public visibility. At the same time, citizens actively participate in political discussions through comments, debates, reactions, and online engagement. These interactions create large-scale user-generated datasets that can be analysed to understand political sentiment, public attention, and influence patterns.

Among social media platforms, YouTube plays an important role in political communication because election-related videos generate substantial public interaction through comments and replies. Political speeches, interviews, debates, campaign discussions, and opinion channels often attract thousands of viewers and create large online discussion spaces where users express support, criticism, humour, anger, and political identity.

This project analyses YouTube discussions related to the Tamil Nadu Election 2026 and compares online public opinion with election outcome data. The project combines Natural Language Processing (NLP), sentiment analysis, topic modelling, temporal analytics, and network analysis to investigate how online political conversations reflect election-related public behaviour.

The analysis focuses on major political parties including TVK, DMK, AIADMK, NTK, BJP, VCK, PMK, DMDK, CPI, CPM, and INC. The project investigates which parties dominated online discussion, how public sentiment varied across political entities, how discussion behaviour changed over time, and how political interaction networks were structured across users, videos, and parties.

Unlike traditional opinion polling, social media analysis captures large-scale real-time public reactions. However, online popularity does not always translate directly into election success. Therefore, this project critically compares online political visibility with actual election outcomes to better understand the relationship between digital engagement and electoral performance.

2. Background and Motivation

The rapid growth of social media has significantly changed the nature of political communication. Digital platforms enable political information to spread instantly across large audiences while also allowing citizens to participate directly in political discussions. Unlike traditional media, social media platforms allow users to react publicly, engage in debates, and repeatedly interact with political content.

YouTube is particularly important for election-related communication because it combines long-form political media content with large-scale public interaction. Political videos often become discussion hubs where users debate election results, leadership quality, party ideology, governance,

campaign promises, and political controversies. These discussions create valuable datasets for understanding online political behaviour.

Tamil Nadu provides a highly suitable context for this analysis because of its strong regional political culture and competitive multi-party environment. The Tamil Nadu Election 2026 generated extensive online engagement involving established parties such as DMK and AIADMK alongside highly discussed political entities such as TVK and NTK. Political leaders including Vijay, Stalin, Seeman, and others were frequently discussed across YouTube election videos.

The motivation for this project is to investigate whether online political engagement reflects broader public opinion and electoral influence. Instead of relying only on basic sentiment classification, the project integrates multiple analytical approaches including NLP, temporal analytics, correlation analysis, topic modelling, and social network analysis. This combined methodology provides deeper insight into both textual political behaviour and structural interaction patterns.

The project also demonstrates how computational social science techniques can be applied to real-world political datasets. By combining election outcomes with social media interaction data, the study provides a broader understanding of digital political ecosystems and modern online political engagement.

3. Project Objectives and Success Criteria

The primary objective of this project is to analyse YouTube political discussions related to the Tamil Nadu Election 2026 and compare online public opinion with election outcomes.

The project aims to analyse political party performance using election result data while simultaneously examining large-scale YouTube discussions related to the Tamil Nadu Election 2026. The study investigates how public sentiment varies across political parties, how political discussion intensity changes over time, and how dominant political themes emerge within online conversations. In addition to textual analysis, the project also focuses on understanding the structural relationships underlying political interaction by constructing party co-occurrence networks, video-party interaction networks, and user-party interaction networks. A further objective of the study is to compare online political visibility and engagement with actual electoral performance to evaluate whether digital discussion patterns meaningfully reflect election outcomes.

The project is considered successful if the final analytical framework integrates NLP and network analysis methods into a coherent and interpretable investigation of online political behaviour. Success is therefore measured not only through technical implementation, but also through the ability of the analysis to produce meaningful insight regarding sentiment, influence, interaction structures, discussion communities, and election-related public opinion.

4. Data Collection

The project used two major datasets: a Tamil Nadu Election 2026 result dataset and a YouTube political discussion dataset.

The election dataset contained constituency-level information including constituency names, candidate names, political parties, total votes, winning candidates, and election-related metrics. This dataset was used to analyse vote distribution, winning margins, and political party performance.

The second dataset was collected using the YouTube Data API. Election-related videos were identified using multiple search queries associated with Tamil Nadu Election 2026 discussions, political interviews, debate programs, election predictions, and political commentary videos.

The YouTube collection process retrieved large-scale election-related metadata including video titles, comment text, replies, likes, timestamps, author names, and reply counts. Collecting both top-level comments and replies was particularly important because it allowed the project to capture broader interaction behaviour rather than analysing isolated comments alone.

The collected dataset was highly suitable for SMNA analysis because it supported both textual analysis and network construction. The comment text enabled sentiment analysis and topic modelling, while user-party relationships, party co-occurrence patterns, and video interactions supported network analysis.

Summary statistics of collected YouTube election discussion dataset

The dataset consisted of 17,730 total comments and replies collected from 385 unique YouTube videos, contributed by 13,762 distinct users. Among these interactions, 12,616 were reply-based discussions, while the preprocessing and cleaning stage produced a final corpus of 5,114 high-quality comments for analysis.

5. Text Preprocessing and Cleaning

The collected YouTube comments contained URLs, punctuation, emojis, repeated characters, multilingual expressions, stopwords, and informal political language. Therefore, extensive preprocessing was required before applying NLP and network analysis techniques.

The preprocessing pipeline included lowercase conversion, URL removal, special character removal, tokenisation, stopword removal, repeated-character handling, and empty comment filtering. Political party names such as TVK, DMK, AIADMK, NTK, BJP, VCK, PMK, and DMDK were intentionally preserved during cleaning because they were central to the analysis.

The preprocessing stage significantly improved the quality of the dataset by reducing noise while retaining politically meaningful terms. This was especially important because many political comments included abbreviations, slogans, party references, and mixed-language expressions.

The final preprocessing pipeline reduced the dataset from over 30,000 raw comments to 22,198 cleaned comments suitable for NLP analysis.

Figure 1. Text preprocessing and analytical workflow pipeline.

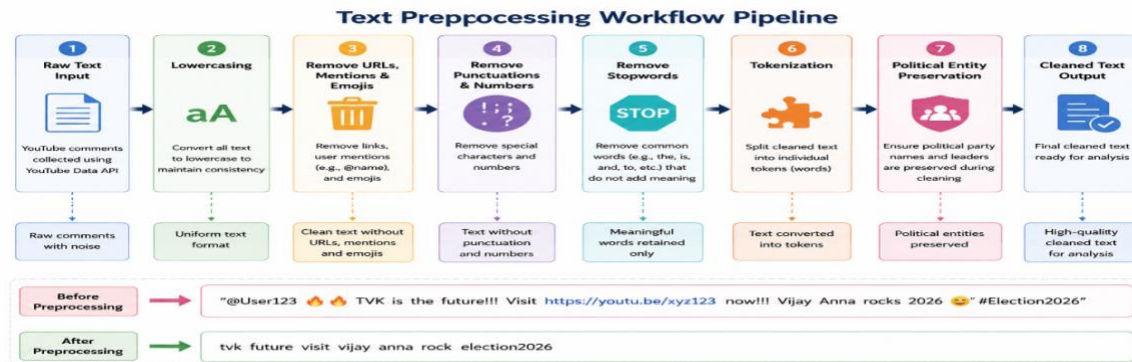


Figure 1. Text preprocessing workflow pipeline.

Example comments before and after preprocessing.

Original: “🔥🔥 TVK is the future!!! Vijay Anna mass 🙌🙌 #Election2026”

Cleaned: tvk future vijay anna mass election2026

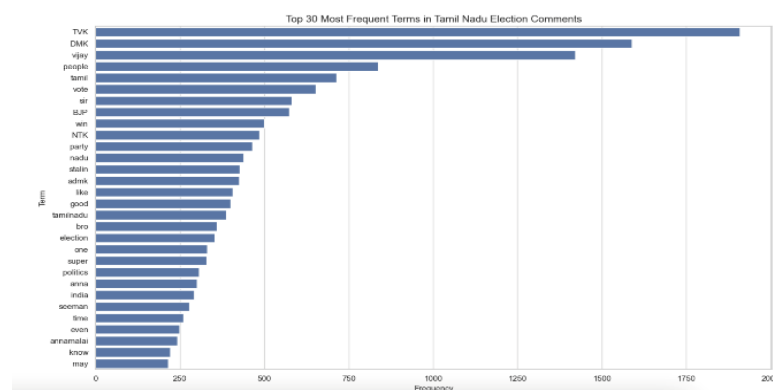
Original: “DMK govt doing good work but BJP media spreading fake news!!!”

Cleaned: dmk govt good work bjp media spreading fake news

6. Public Opinion and Political Discussion Analysis

Exploratory text analysis was performed to identify dominant political discussion patterns within the YouTube dataset. The analysis examined frequent terms, party mentions, political phrases, and overall public discussion behaviour.

Figure 2. Top 30 Most Frequent Terms in Tamil Nadu Election Comments

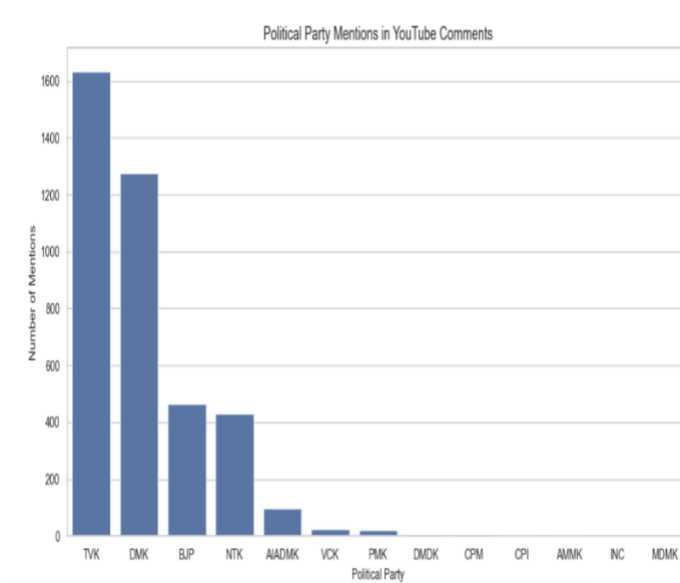


The frequency analysis also showed repeated discussion of election-related terms associated with leadership, voting, seats, public support, and campaign narratives. This confirms that the collected dataset remained strongly aligned with election-related public discussion.

The results showed that online political discussion was heavily concentrated around a limited set of major political parties. TVK, DMK, NTK, BJP, and AIADMK appeared most frequently in

YouTube discussions, indicating that these parties dominated public online attention during the election period.

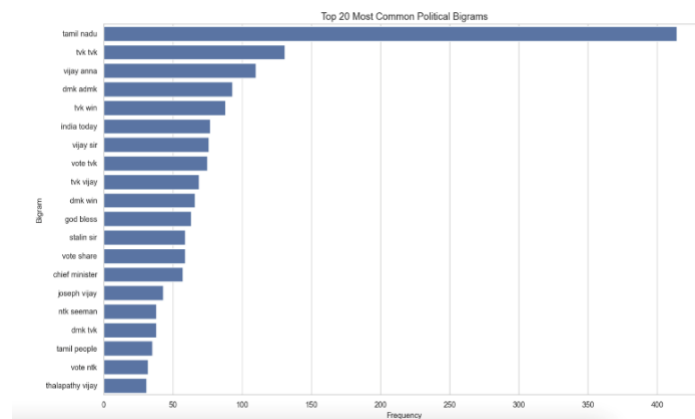
Figure 3. Political Party Mentions in YouTube Comments



The party mention analysis revealed that TVK received the highest level of online attention within the dataset. DMK and NTK also generated substantial discussion volume, while BJP and AIADMK maintained strong visibility across election-related conversations.

The dominance of these parties suggests that online political engagement was concentrated around highly competitive and highly visible political actors.

Figure 4. Top 20 Most Common Political Bigrams



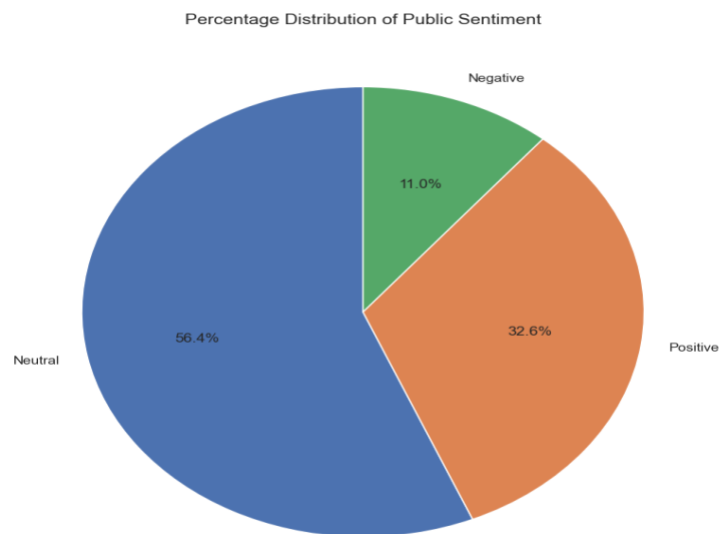
The bigram analysis provided deeper insight into political discussion structure by identifying repeated two-word combinations associated with parties, leaders, election narratives, and campaign-related expressions.

7. Sentiment Analysis

Sentiment analysis was performed using the VADER sentiment analysis model. VADER was selected because it is specifically designed for social media text and performs effectively on short informal comments.

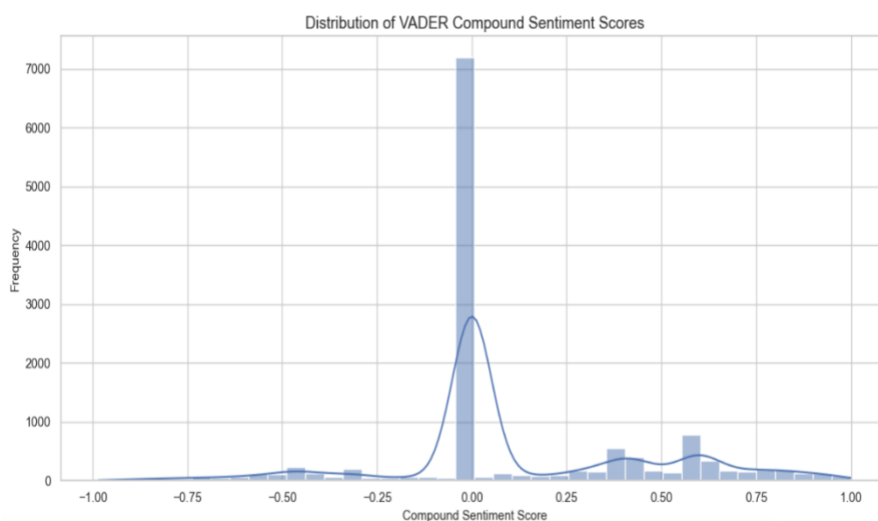
Each cleaned YouTube comment received negative, neutral, positive, and compound sentiment scores. The compound score was then used to classify comments into positive, neutral, and negative sentiment categories.

Figure 5. Overall sentiment distribution statistics.



Final sentiment distribution showed that neutral sentiment dominated political discussions, followed by positive sentiment, while negative sentiment formed the smallest category. The prevalence of neutral sentiment suggests that many comments involved factual observations, political reactions, or short conversational expressions rather than strongly emotional language.

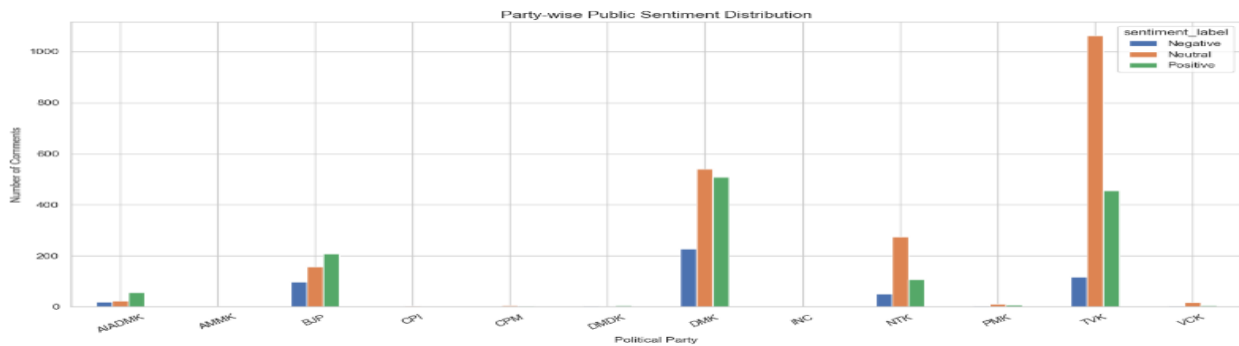
Figure 6. Distribution of VADER compound sentiment scores



The distribution of VADER compound sentiment scores revealed a strong concentration of comments around the neutral sentiment range, with the highest peak occurring near a compound score of 0. This indicates that a large proportion of YouTube election discussions were

neutral or moderately opinion-based rather than strongly emotional. However, the graph also shows noticeable positive sentiment peaks between 0.4 and 0.7, suggesting the presence of supportive political discussions and favourable public reactions towards certain parties or leaders. In contrast, strongly negative sentiment values were comparatively limited, indicating that highly critical or aggressive political reactions formed a smaller portion of the overall discussion dataset.

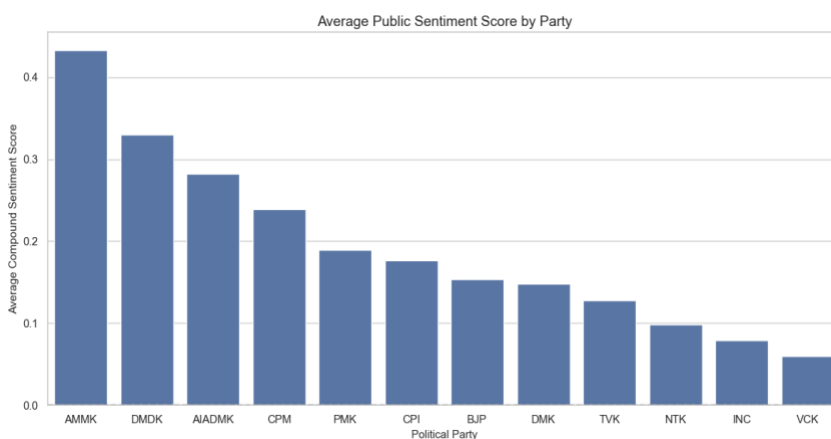
Figure 7. Party-wise sentiment Distribution



Party-wise sentiment analysis showed important differences between political entities. While TVK and DMK generated the highest overall discussion volume, the sentiment associated with these parties was mixed rather than uniformly positive.

This is an important finding because online visibility alone should not automatically be interpreted as online support. Political parties may receive attention because of popularity, criticism, controversy, or repeated media visibility.

Figure 8. Avg sentiment score by political party

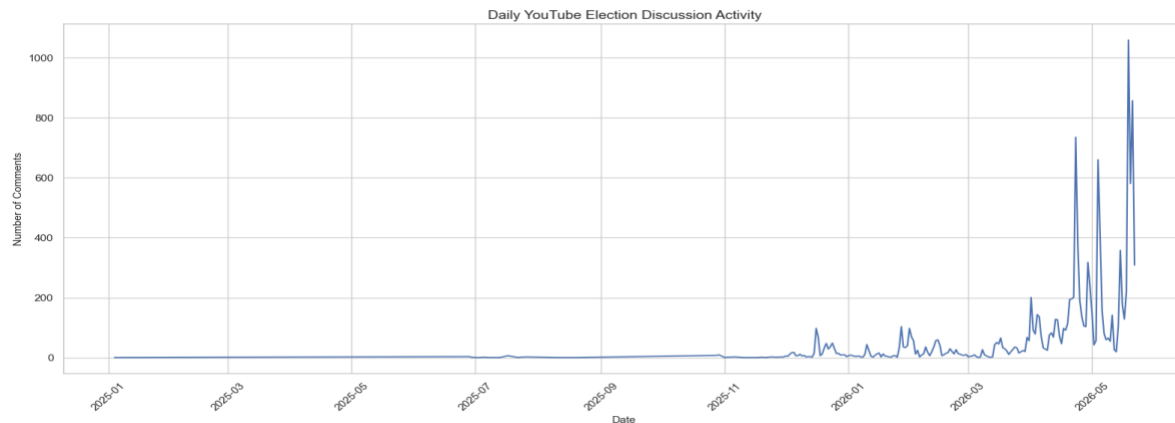


The average sentiment analysis showed that some parties with lower discussion volume still achieved relatively strong average sentiment values. This demonstrates that sentiment intensity and discussion popularity are not always directly correlated.

8. Temporal Public Discussion Analysis

Temporal analysis was conducted to understand how political discussion activity changed throughout the election period. Timestamp-based analysis was used to examine daily discussion activity, sentiment variation over time, and high-engagement political discussion periods.

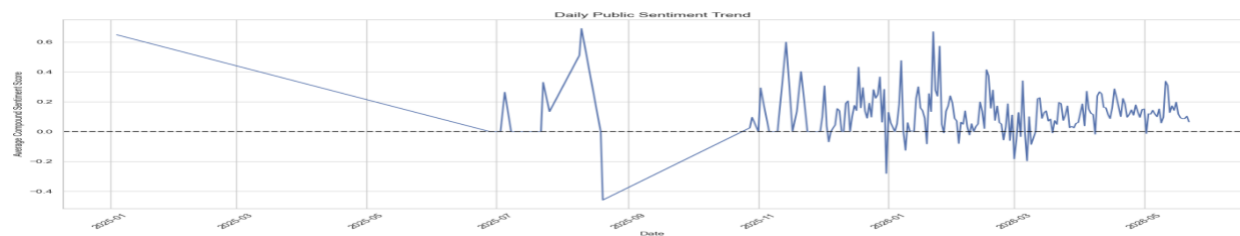
Figure 9. Daily political discussion activity timeline.



The results showed that political discussion intensity varied significantly across time rather than remaining evenly distributed. Certain dates experienced major spikes in political engagement, likely associated with campaign events, interviews, election predictions, or viral political content.

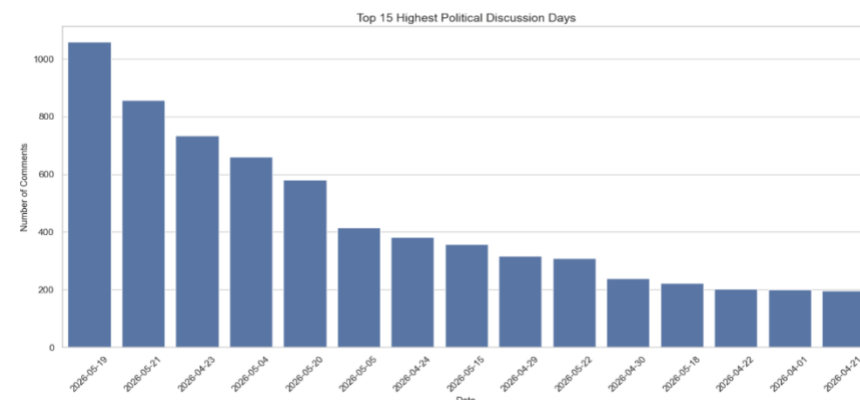
The daily activity graph demonstrated that online political attention became concentrated around specific high-engagement periods. This behaviour is consistent with event-driven political communication patterns commonly observed on social media platforms.

Figure 10. Daily public sentiment trend over time



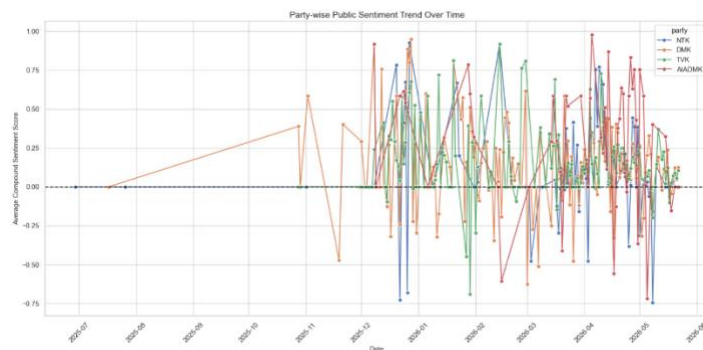
The sentiment trend analysis showed fluctuations in emotional tone across the election discussion period. Some dates showed increased positive discussion intensity, while others reflected stronger neutral or mixed reactions.

Figure 11. Most active political discussion days



The most active discussion days corresponded to periods of heightened political visibility and major election-related public engagement.

Figure 12. Party-wise sentiment timeline

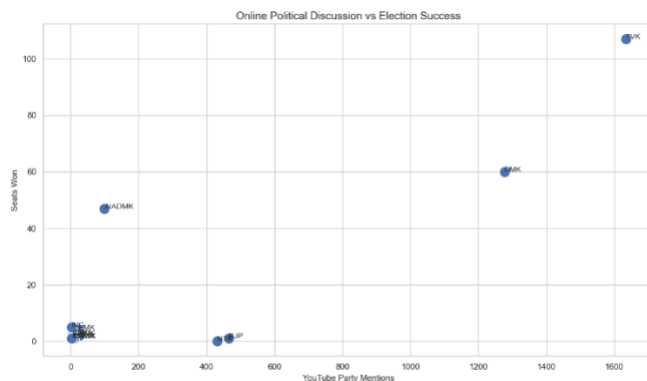


The party sentiment timeline revealed that public sentiment towards major parties changed over time rather than remaining constant. This suggests that online public perception was influenced by evolving political events, campaign narratives, and media exposure.

9. Election Comparison and Correlation Analysis

A key objective of this project was to compare online political popularity with real election outcomes. Party mention frequency from YouTube discussions was therefore compared with election performance metrics including seats won.

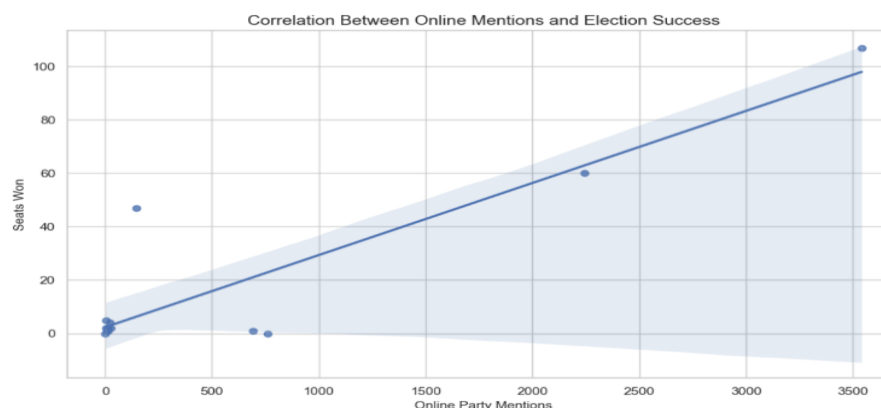
Figure 13. Online political mentions versus seats won.



The comparison analysis revealed a strong positive relationship between online political visibility and electoral performance within the Tamil Nadu Election 2026 dataset. Major political parties such as TVK and DMK achieved both high discussion intensity across YouTube conversations and strong election outcomes. However, the relationship was not perfectly proportional, as some

parties generated substantial online attention without equivalent electoral success. These findings suggest that social media discussions can reflect broader public engagement and political visibility, although online popularity alone should not be interpreted as a direct predictor of voting behaviour.

Figure 14. Correlation between online mentions and election performance.



The correlation analysis produced a strong positive correlation value, indicating that online discussion intensity and election outcomes were meaningfully related within the dataset.

Nevertheless, the differences observed between discussion volume and seat outcomes highlight the importance of interpreting social media metrics carefully.

10. Topic Modelling Analysis

Topic modelling was performed using Latent Dirichlet Allocation (LDA) to identify hidden thematic structures within the YouTube comments. Unlike simple frequency analysis, topic modelling groups related words into broader discussion themes.

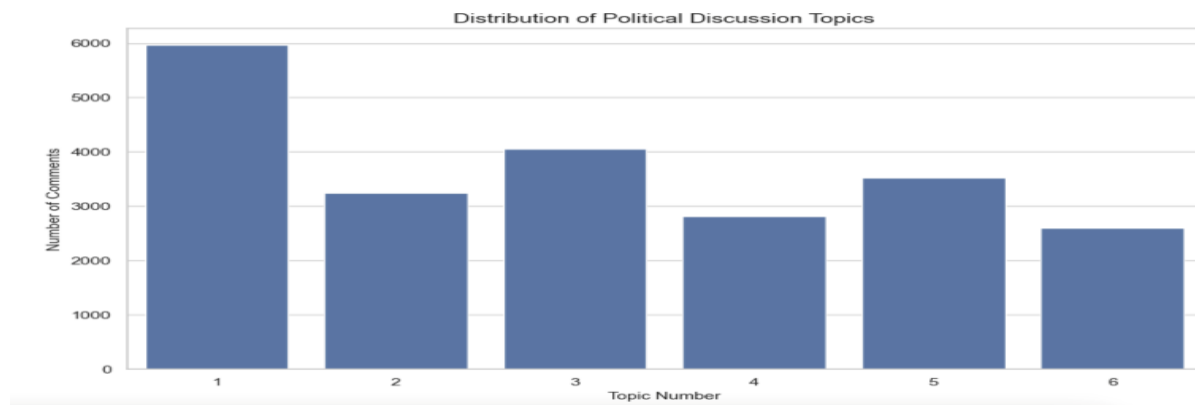
The analysis identified six major political discussion topics involving election competition, leadership, party-related narratives, voting behaviour, public support, and campaign discussion. The extracted topics demonstrated that online political discussion was highly diverse and multi-dimensional rather than concentrated around a single political issue. The identified themes revealed strong public focus on political leadership, election competition, governance narratives, campaign strategies, public support, and party-based political rivalry. This indicates that YouTube discussions captured a broad range of political concerns and reflected active public engagement across multiple dimensions of the Tamil Nadu election ecosystem.

Topic	Top keywords
NTK and Leadership Discussion	NTK, Seeman, party, India, people, Tamil Nadu, leader, vote, media
TVK and Vijay Support Discussions	TVK, Vijay, vote, win, Thalapathy, dreams, thanks, anna
AIADMK and Informal Political Conversations	AIADMK, Vijay, love, correct, bro, unga, poda
DMK and Election Outcome Discussions	DMK, AIADMK, win, true, welcome, Ram, Saravanan
BJP, Annamalai and Governance Discussions	BJP, Annamalai, Tamil Nadu, DMK, government, seats, people
Stalin, Media and Political Reactions	Keywords: Stalin, Vijay, news, fake, survey, Modi, super, anna, joke

Distribution of extracted political discussion topics

The topic distribution analysis showed that certain topics dominated online political discussion more strongly than others. Topics associated with TVK, DMK, elections, leadership, and campaign-related narratives appeared most frequently.

Figure 15. Distribution of extracted political discussion topics



11. Political Network Analysis

Political network analysis formed the most important postgraduate-level component of this project because it examined the structural relationships underlying online political discussions.

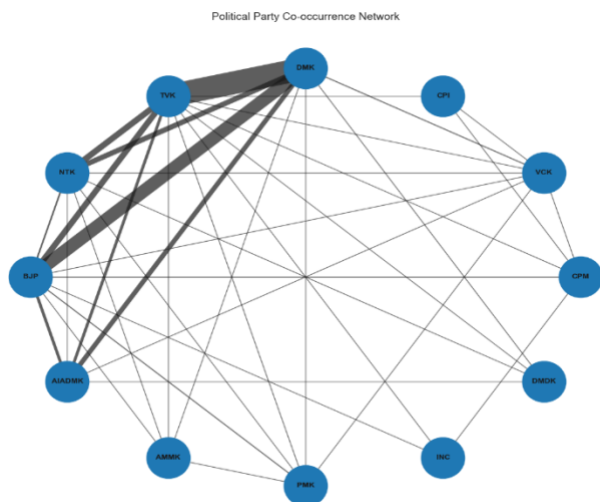
Three major network structures were analysed within the project. The first was a political party co-occurrence network designed to identify relationships between parties mentioned together within YouTube discussions. The second was a video–party interaction network that connected political parties with election-related videos generating large-scale public engagement. The third was a user–party interaction network that examined how politically active users clustered around different parties and discussion communities.

The political interaction networks were constructed using the Python NetworkX library, which enabled graph construction, centrality analysis, and community-based structural investigation. Network visualisations were further refined using Gephi to improve interpretability and provide clearer visual representations of political interaction patterns and community structures.

In the party co-occurrence network, nodes represented political parties while edges represented parties being mentioned together within the same comments. The network was constructed as an undirected weighted graph because co-mentioned political parties represent mutual relationships rather than one-way interactions. Edge weights represented the frequency of co-occurrence between political parties across YouTube discussions, allowing the analysis to capture both the existence and strength of political interaction relationships.

The party co-occurrence network revealed that parties such as TVK, DMK, BJP, and AIADMK occupied highly central positions within the online political discussion structure.

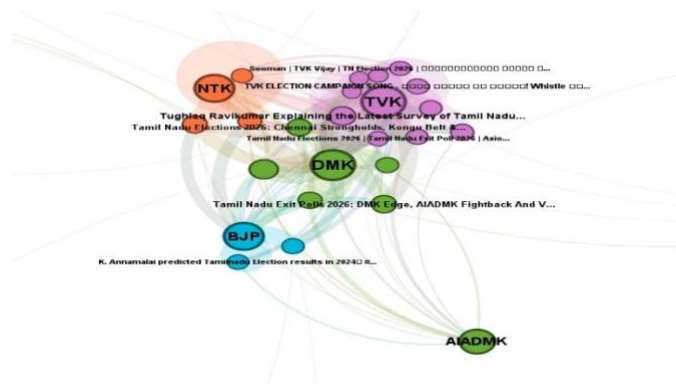
Figure 16. Political party co-occurrence network.



The network structure demonstrated strong inter-party connectivity, indicating that YouTube discussions frequently compared or linked multiple parties within the same political conversations.

The video-party interaction network connected election-related videos with political parties discussed within those videos. This network structure revealed how certain videos acted as highly influential political discussion hubs.

Figure 17. Video-party interaction network

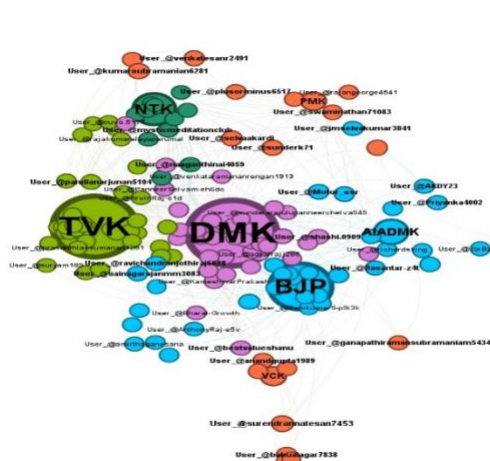


The video-party network showed that highly visible election videos frequently generated discussions involving multiple political parties simultaneously. TVK, DMK, BJP, and AIADMK occupied major positions within this network.

The user-party interaction network connected politically active users with the parties they discussed. This network

revealed audience clustering behaviour and online political interaction communities.

Figure 18. User-party interaction network

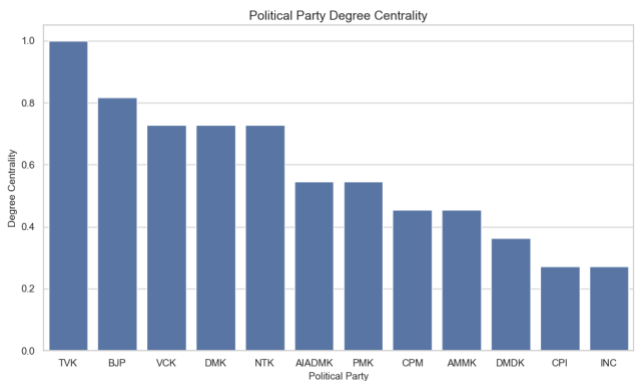


The user-party interaction network revealed the presence of clear party-centred discussion communities within the YouTube political ecosystem. These communities indicate that politically active users tended to cluster around particular parties and political narratives. However, substantial overlap between multiple discussion groups was also observed, suggesting that political interaction on YouTube was not completely fragmented

into isolated echo chambers. This overlapping structure indicates continued exposure to competing political viewpoints and cross-party discussion behaviour within the online election environment.

Centrality analysis was also performed to measure influence and structural importance within the political interaction network.

Figure 19. Degree centrality values for major political parties.



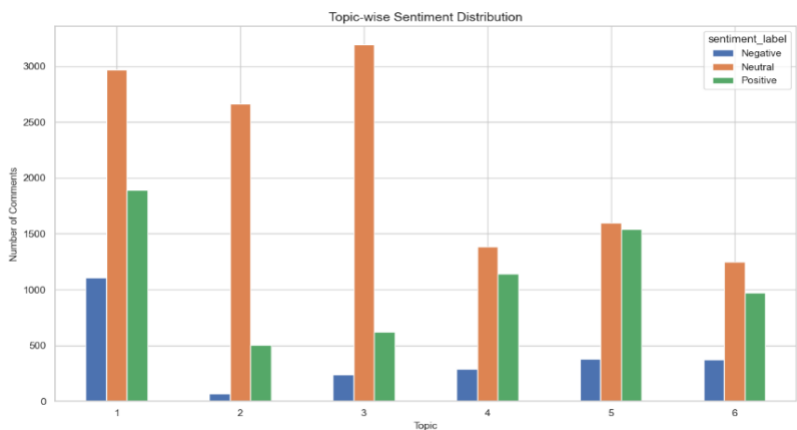
The centrality analysis demonstrated that major political parties such as TVK, DMK, BJP, and AIADMK occupied highly connected positions within the political interaction network. High degree centrality values indicate that these parties were frequently associated with multiple political entities across YouTube discussions, suggesting their dominant structural influence

within the online political communication ecosystem. The results further demonstrate that political discussions were heavily concentrated around a relatively small number of highly influential parties.

12. Advanced Political Analytics

The advanced political analytics section combined topic modelling, sentiment analysis, and engagement behaviour to examine deeper political interaction patterns within the YouTube election discussion environment. This stage focused on understanding how political parties, public sentiment, discussion themes, and engagement metrics interacted across online political conversations.

Figure 20. Topic-wise sentiment distribution across Tamil Nadu Election 2026 discussions.



The topic-wise sentiment analysis showed that different political discussion themes generated different emotional responses. Some topics were dominated by neutral discussion, while others attracted stronger positive or negative reactions, indicating variation in public engagement across political narratives.

Party versus Political Discussion Topics

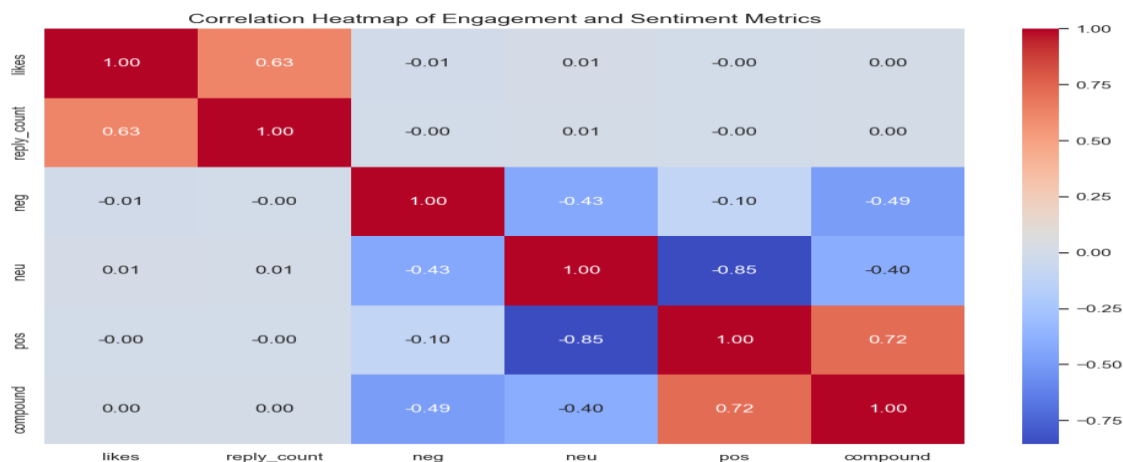
Political Party	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6
AIADMK	45	0	2	5	84	8
AMMK	1	0	0	0	5	0
BJP	269	9	28	70	245	72
CPI	0	0	0	0	4	0
CPM	0	0	4	0	7	0
DMDK	7	0	0	0	4	0
DMK	537	40	678	283	567	137
INC	0	0	1	1	2	0
MDMK	0	0	1	0	0	0
NTK	125	12	56	32	519	16
PMK	6	0	0	0	7	7
TVK	409	2161	272	189	366	144
VCK	8	0	5	0	11	0

The party-topic association analysis demonstrated that parties were not discussed uniformly across all themes. TVK showed extremely strong concentration within Topic 6, while DMK maintained high visibility across multiple topics, particularly Topics 2 and 3. BJP and NTK also appeared across several discussion themes, reflecting broader political visibility within online conversations.

[illegible]

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Figure 23. Correlation heatmap of engagement metrics and sentiment variables



The correlation analysis showed that likes and replies were moderately positively related, while sentiment polarity had relatively weaker direct relationships with engagement metrics. This suggests that online political engagement was influenced not only by emotional tone, but also by visibility, controversy, and political relevance.

Average Likes by Sentiment Category

- **Neutral comments** received the highest average engagement with **8.95 likes**.
- **Positive comments** received an average of **7.35 likes**.
- **Negative comments** received the lowest average engagement with **4.14 likes**.

Overall, the advanced analytics section demonstrated that online political influence emerged through interconnected relationships involving political themes, sentiment behaviour, public engagement, and party visibility rather than through isolated metrics alone.

13. Discussion

The combined findings demonstrate that YouTube political discussions surrounding the Tamil Nadu Election 2026 were strongly concentrated around a small number of dominant political parties. TVK and DMK consistently appeared as major discussion hubs across frequency analysis, sentiment analysis, temporal analysis, and network structures.

The integration of NLP analysis and social network analysis revealed that online political influence extended beyond simple sentiment polarity alone. Political parties that occupied highly central positions within the interaction networks also tended to dominate public discussion topics, sentiment visibility, and engagement intensity. This suggests that structural prominence within online political communication networks may play an important role in shaping discussion visibility, information flow, and public attention during election periods.

The sentiment analysis showed that online political discussions were largely neutral in tone, with positive sentiment significantly exceeding negative sentiment. However, party-level sentiment analysis demonstrated that discussion popularity and positive sentiment were not always directly aligned.

The election comparison analysis produced one of the strongest findings of the project by showing a strong positive relationship between online mentions and election performance. This suggests that YouTube political visibility partially reflected real-world electoral strength.

Nevertheless, differences between online visibility and electoral success also highlighted the limitations of social media metrics. Some parties generated substantial discussion without proportional election outcomes, indicating that online attention may reflect controversy, celebrity influence, or media exposure rather than direct voter conversion.

The topic modelling analysis revealed that political discussions were multi-dimensional and involved leadership, elections, party competition, governance, and public opinion themes.

The network analysis provided the strongest structural insight. The party co-occurrence network showed strong inter-party discussion overlap, while the video-party network demonstrated the importance of influential media content in shaping online political interaction. The user-party interaction network further revealed clustered audience behaviour and overlapping political discussion communities.

Together, these findings demonstrate that online political communication operates as a complex interaction ecosystem involving users, parties, sentiment, media content, and network structures.

14. Limitations

Although the project produced meaningful analytical findings, several limitations should be acknowledged.

First, the analysis focused only on YouTube comments and therefore may not represent the entire voting population of Tamil Nadu. Social media users may differ significantly from the general electorate in terms of demographics, digital access, and political engagement.

Second, the YouTube API collection process depends on search queries, available videos, comment visibility, and API restrictions. Therefore, the collected dataset represents a large sample of public discussion rather than a complete record of political opinion.

Third, VADER sentiment analysis may not fully capture sarcasm, political humour, Tamil-English mixed expressions, slang, or context-specific meanings present within election discussions.

Another limitation of the study is that the constructed political interaction networks represent patterns of online discussion and co-occurrence rather than real-world political alliances or direct interpersonal relationships. The identified network structures therefore reflect discussion behaviour within the YouTube platform rather than verified political coordination between parties

or individuals. Consequently, network influence should be interpreted as online visibility and interaction prominence rather than direct real-world political influence.

Finally, online political attention does not necessarily represent direct electoral support. Political parties may receive substantial discussion volume because of controversy, criticism, celebrity influence, or repeated media visibility.

15. Conclusion

This project analysed YouTube political discussions related to the Tamil Nadu Election 2026 using a combined NLP and network analysis approach.

The final analysis pipeline successfully integrated election result analysis, YouTube data collection, text preprocessing, sentiment analysis, temporal analytics, topic modelling, correlation analysis, and political network analysis.

The findings demonstrated that TVK, DMK, NTK, BJP, and AIADMK dominated online political discussions. Sentiment analysis revealed that neutral discussion was most common, while party-wise analysis showed meaningful differences in public sentiment patterns.

The election comparison analysis showed a strong relationship between online political visibility and election performance, although online discussion intensity did not perfectly predict electoral success.

The topic modelling results revealed multiple political discussion themes involving leadership, voting, campaign narratives, and political competition.

Most importantly, the network analysis demonstrated that online political discussions formed interconnected interaction ecosystems involving users, parties, and election-related media content.

Overall, the project demonstrates how computational social science techniques combining NLP, temporal analytics, correlation analysis, and social network analysis can provide meaningful insight into modern political communication ecosystems. The findings show that large-scale social media discussions contain valuable information regarding public attention, political visibility, interaction structures, and online engagement behaviour during election periods. The study further highlights the importance of integrating both textual and structural analytical approaches when investigating digital political environments.

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